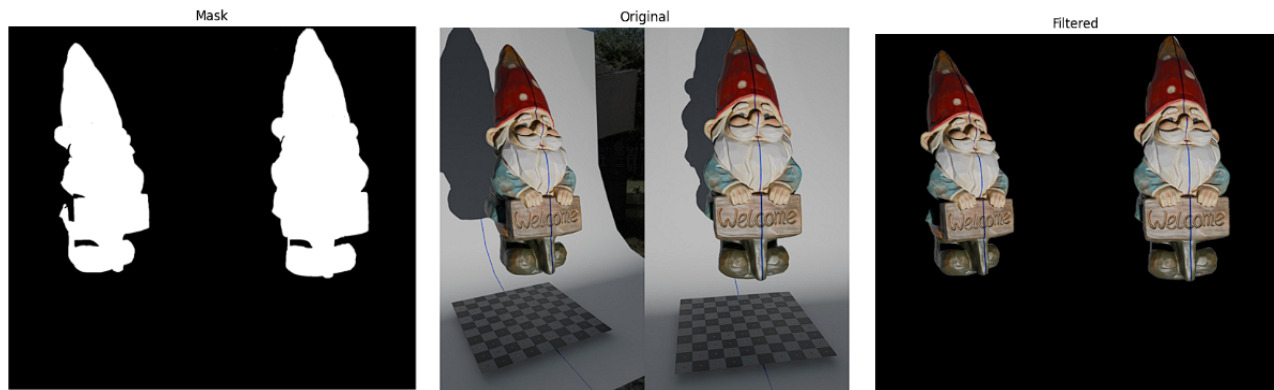


Final Project – Report

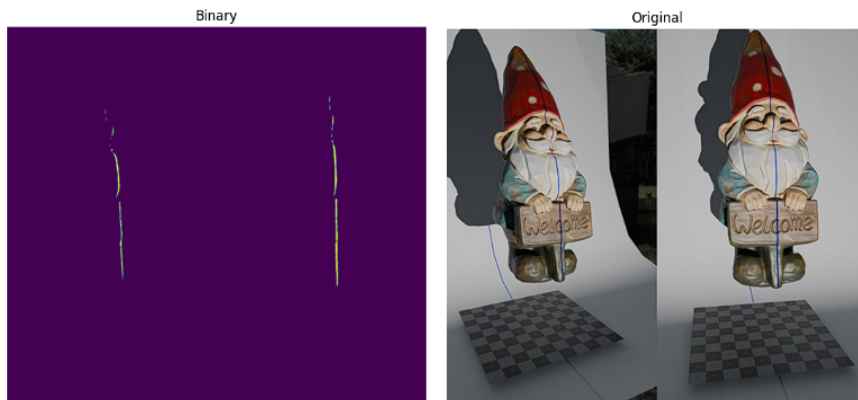
Student = David Medina Rosner

Id = F11115117

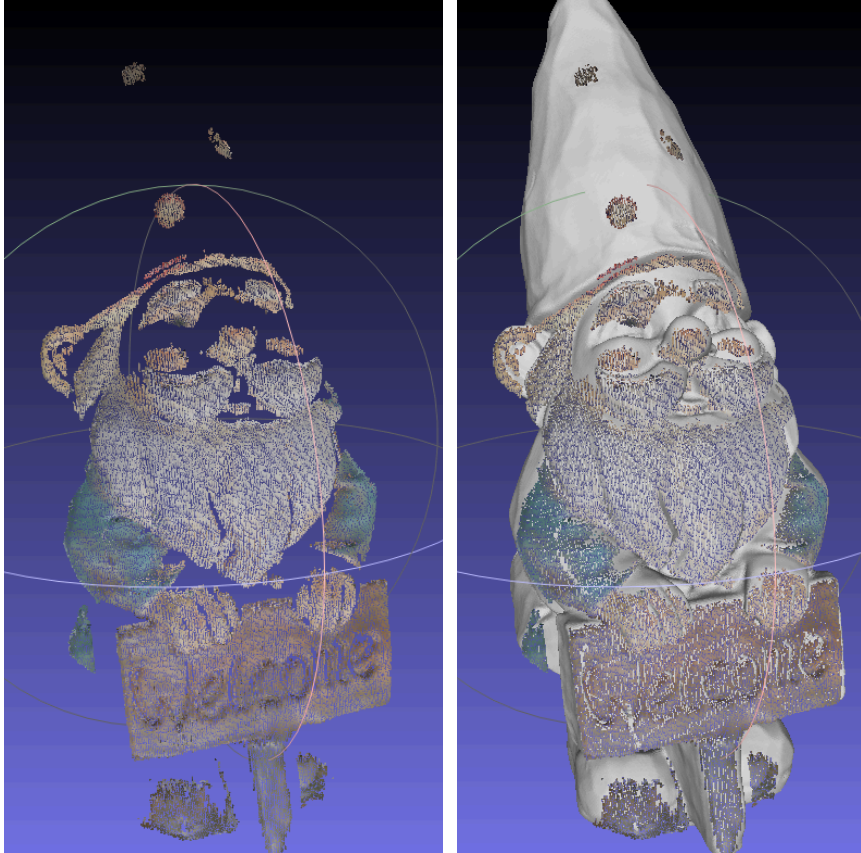
For this project of “Reconstruct 3D from stereoscopic side-by-side images”, first, I create a mask of Santa's “000.jpg” image. This mask was created using OpenCV GrabCut algorithm for image segmentation. With this mask I am able to filter all background noise, and can focus only on what is projected onto Santa itself.



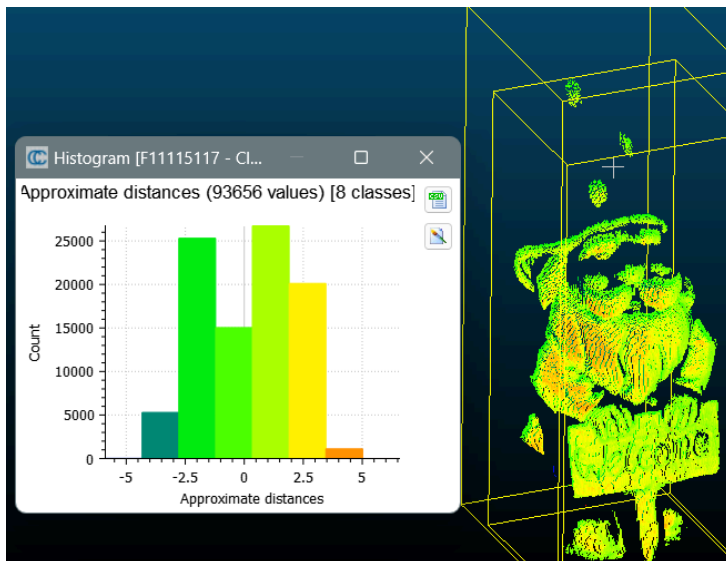
By using manipulating the RGB channels of the image, I filter the image for when the blue channel is greater than the red and green channels by a threshold of 20. This way I get a binary image for each picture of the blue line laser projected into Santa. This method is fairly accurate to estimate Santa's 3D points of the body, however, that's not the case for Santa's red hat. Because when the blue line laser is projected onto the red hat, the resulting color is a deep blue very similar to black.



Bellow is the result of the 3D points generated by calculating the epipolar lines, and performing 3D Direct Triangulation method, as mentioned in the lectures. Only points below an error threshold of 150mm are included in the 3D point list. Decreasing the threshold causes distortion of the projection.



From CloudCompare, I get that there are some outliers that are about 5.73mm apart. However, on average I get that most points are about 1.3mm apart.



1	Min dist.	0
2	Max dist.	5.73091
3	Avg dist.	1.37015
4	Sigma	0.901329
5	Max error	0.694975